

NIRUDU MACHAGIRI

Software Engineer | Python Developer | AI, Data & Cybersecurity Enthusiast
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PROFESSIONAL SUMMARY

Motivated final-year B.Tech student with strong foundations in Python, backend development, and cybersecurity. Certified in Data Science (Infosys Springboard), Oracle Cloud AI Foundations, and JP Morgan Software Engineering Simulation. Experienced in building full-stack web applications, data pipelines, intrusion detection systems, and REST APIs. Eager to contribute to AI-driven and security-aware engineering roles.

TECHNICAL SKILLS

Languages: Python, SQL, C, JavaScript

Frameworks & Libraries: Django, React.js, Redux, Flask, Pandas, NumPy, Scikit-learn

AI / Data Skills: Data Preprocessing, Text Cleaning, LLM Fine-tuning, Prompt Engineering

Cybersecurity: Intrusion Detection, Vulnerability Assessment, Penetration Testing Basics, OWASP Top 10, Secure Coding, Log Analysis, Firewall & IDS/IPS, Wireshark, Nmap

Tools & Platforms: Git, GitHub, VS Code, MySQL, PostgreSQL, Docker, Jupyter Notebook, Oracle Cloud, Postman

Concepts: OOP, REST APIs, Data Structures, Algorithms, SDLC, Agile Basics

EDUCATION

B.Tech – Computer Science & Engineering in Data Science 2023 – 2026

TKR College of Engineering and Technology, Hyderabad – CGPA: 6.6

Diploma – Electronics & Communication Engineering 2020 – 2023

Teegala Krishna Reddy Engineering College, Hyderabad – Score: 6.19

CERTIFICATIONS

- Data Science – Infosys Springboard (2024): Python for Data Science, Data Analysis, ML fundamentals, hands-on real-world projects.
- Oracle Cloud Infrastructure AI Foundations – Oracle (2024): AI/ML services on OCI, Generative AI, LLM deployment concepts, and cloud-based AI infrastructure.
- Software Engineering Job Simulation – JP Morgan Forage (2024): Patched broken repositories, local dev environment setup, used Perspective library for financial data visualization.

PROJECTS

Zepto Clone – Full-Stack Quick-Commerce App Django | React.js | PostgreSQL | Redux

- Built a fully functional quick-commerce web app inspired by Zepto with product browsing, cart management, order placement, and real-time inventory updates.
- Developed a Django REST Framework backend with JWT-based authentication, role-based access control for admin/user flows, and optimized PostgreSQL database queries.
- Designed a responsive React.js frontend with Redux for global state management, dynamic product filtering, search, and seamless checkout flow with payment simulation.

Intrusion Detection System – B.Tech Major Project Python | Scikit-learn | Networking

- Designed ML-based IDS detecting DoS, probe, and R2L attacks using Random Forest & SVM on NSL-KDD dataset, achieving 97%+ accuracy with minimal false positives.
- Built real-time packet monitoring using Python socket library and Wireshark for live traffic capture, alert generation, and security log analysis.
- Applied anomaly detection, traffic classification, and secure coding practices to minimize attack surface and improve system resilience.

KEY STRENGTHS

Strong Problem-Solving | Fast Learner | Full-Stack Development | Security-Aware Development |
Attention to Data Quality | Adaptable to Rotational Shifts (Day WFO / Night WFH) | Team Collaboration